Deep Learning and Neural Network

大作业：训练一个神经网络

2026-09-21

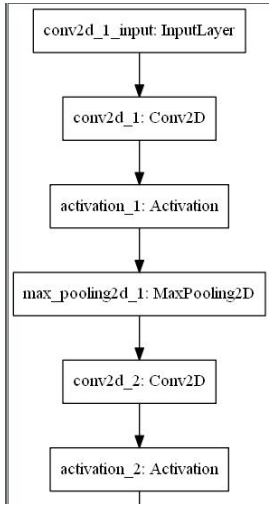
- 作业介绍
  - 训练一个神经网络模型，数据类型可以为图像、时间序列数据、音频、文本等等。
  - 形成项目报告。

• 提交要求

- 项目报告分为三个部分来写，模型设计、训练效果、测试结果。

必须包含以下几点：

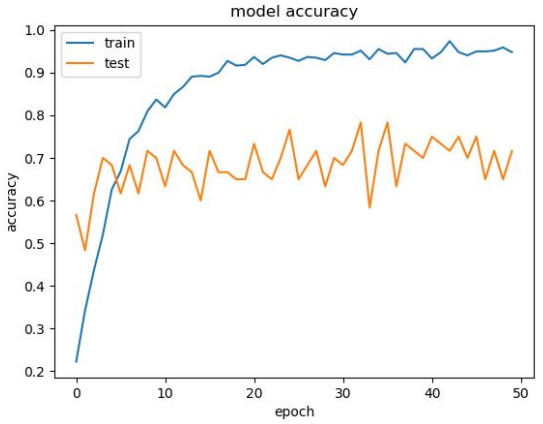
✓网络模型结构，例：

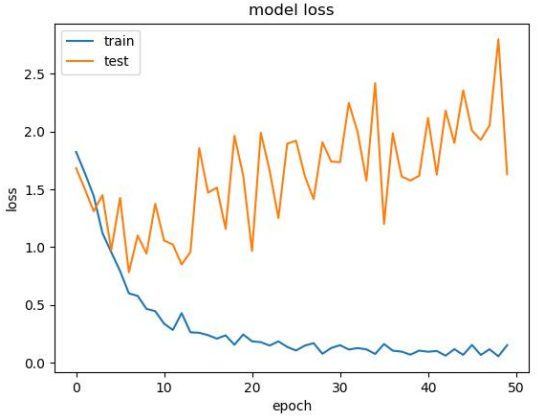


✓实验结果图，包含accuracy和loss的曲线。例：

✓模型识别的准确率

提交训练集的准确率与在验证集上的准确率



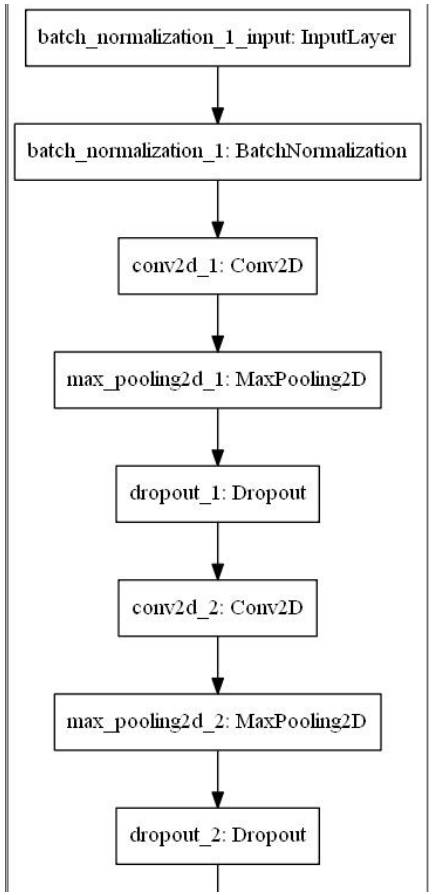


- 实验环境
  - Pytorch框架
  - IDE不限，Jupyter notebook 、Pycharm 等

- 参考

```
#网络模型展示
model.summary()
plot_model(model, to_file='your/project/path/model.png')
```

```
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
batch_normalization_1 (Batch (None, 224, 224, 3)       12
_________________________________________________________________
conv2d_1 (Conv2D)            (None, 224, 224, 64)      3136
_________________________________________________________________
max_pooling2d_1 (MaxPooling2 (None, 112, 112, 64)      0
_________________________________________________________________
dropout_1 (Dropout)          (None, 112, 112, 64)      0
_________________________________________________________________
conv2d_2 (Conv2D)            (None, 109, 109, 64)      65600
_________________________________________________________________
max_pooling2d_2 (MaxPooling2 (None, 54, 54, 64)        0
_________________________________________________________________
dropout_2 (Dropout)          (None, 54, 54, 64)        0
_________________________________________________________________
flatten_1 (Flatten)          (None, 186624)            0
_________________________________________________________________
dense_1 (Dense)              (None, 256)               47776000
_________________________________________________________________
dropout_3 (Dropout)          (None, 256)               0
_________________________________________________________________
dense_2 (Dense)              (None, 128)               32896
_________________________________________________________________
batch_normalization_2 (Batch (None, 128)               512
_________________________________________________________________
dense_3 (Dense)              (None, 100)               12900
=================================================================
Total params: 47,891,056
Trainable params: 47,890,794
Non-trainable params: 262
```



注意事项

- ✓ 不要在报告中出现科普性介绍、如介绍卷积层、激活函数。报告需规范学术语言，尽量简洁，尽量控制在10页以内。
- ✓ 希望大家尝试设计新的网络模型解决各种任务，学习调参方式、以及一些过拟合的抑制方式。
- ✓ 若能在实验报告中体现出数据处理的方式、模型设计的原则、调参的方法等内容，会考虑加分。